

Software Requirements Specification (SRS)

Hostel Management System

Project: Hostel Management System

Version: 1.0

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Version	Date	Author	Change summary	Approval
1.0	05-09-2026	Project Team	Initial SRS for Hostel Management System based on assigned project scope	For coordinator review

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

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1. Introduction

1.1 Purpose

This Software Requirements Specification defines the functional, non-functional, interface, security, quality, acceptance, and traceability requirements of the Hostel Management System (HMS). The HMS is intended to

manage hostel room allocation and student details through a web-based application. This document serves as the requirements baseline for design, implementation, testing, and evaluation of the project.

1.2 Scope

The Hostel Management System provides a centralized means to maintain student records, hostel and room records, room availability, room allocation, room transfer/vacation, and allocation history. The primary purpose is to simplify and control the process of maintaining student details and assigning students to hostel rooms while preserving accurate allocation information.

The system excludes physical hostel construction or maintenance, biometric hardware control, external institutional ERP integration, payment processing, food/mess management, attendance management, and other hostel services not required to manage student details and room allocation.

1.3 Audience

- Students/residents who need to view their own student and room information.
- Wardens/hostel staff responsible for maintaining student, hostel, room, and allocation records.
- Administrators responsible for system-wide user, student, hostel, room, and allocation management.
- Developers and testers implementing and verifying the system.
- Course instructors and assessment evaluators reviewing project completeness.

1.4 Definitions and Acronyms

Term	Definition
HMS	Hostel Management System
FR	Functional Requirement
NFR	Non-Functional Requirement
RBAC	Role-Based Access Control
UML	Unified Modeling Language
RTM	Requirements Traceability Matrix
UI	User Interface
DB	Database
ID	Identifier

2. Overall description

2.1 Product perspective

The HMS is a web-based client-server application. Students, wardens, and administrators access the system through a web browser. The application layer performs authentication, authorization, student-record management, hostel/room management, allocation processing, and reporting, while a relational database stores persistent

records. The implementation shall use one of the permitted technology stacks assigned for this project: Python with Django or Flask, or PHP with Laravel.

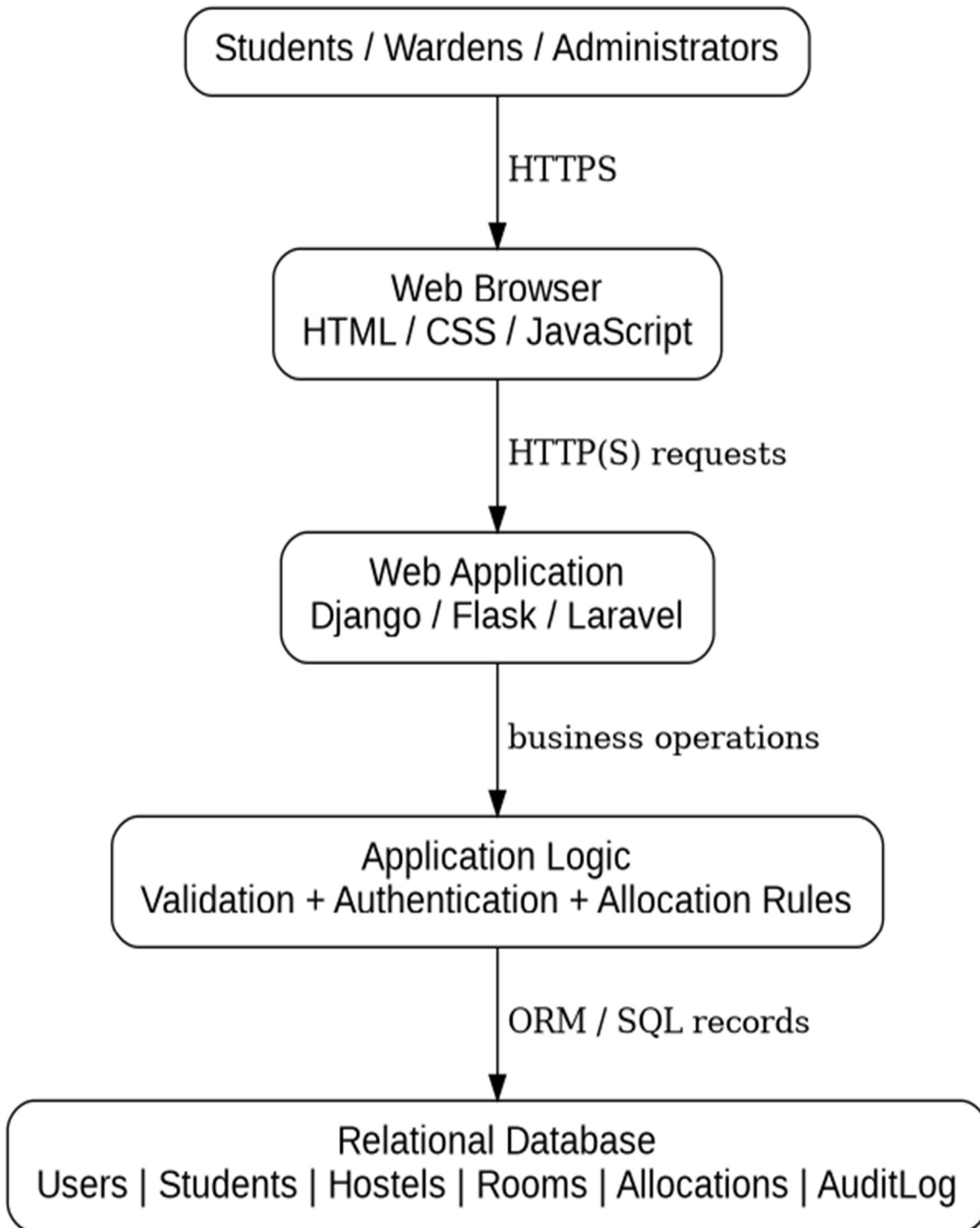


Figure 1. High-level HMS architecture

2.2 Major product functions (detailed)

- Authenticate users and enforce role-based access according to role.
- Register and maintain student details.

- Search and view student records for authorized users.
- Create and maintain hostel records.
- Create and maintain room records and capacity information.
- Display current room availability and occupancy.
- Allocate eligible students to available hostel rooms.
- Prevent invalid allocations and room over-capacity.
- Transfer students between rooms while preserving allocation history.
- Vacate active room allocations and record the end date.
- Allow students to view their own current room/allocation details.
- Allow authorized staff to view allocation history.
- Provide allocation and occupancy summaries for authorized staff.
- Maintain audit information for critical student, room, and allocation changes.
- Allow authorized administrators to manage user roles and access.
- Provide validation and controlled error handling for invalid operations.

2.3 User roles and characteristics

Role	Characteristics	Primary capabilities
Student	Basic computer/web knowledge; accesses own records	Login, profile, room details, allocation status, room availability
Warden	Operational hostel staff; manages one or more hostels	Students, rooms, allocations, transfers, vacancy, dashboards, reports
Administrator	System-wide authorized user	Users/roles, hostels, rooms, students, allocations, reports, audits

2.4 Operating environment

- Client: current Chrome, Edge, or Firefox browser on a desktop/laptop or modern mobile device.
- Application: one of the permitted stacks — Python with Django or Flask, or PHP with Laravel.
- Database: a relational database supported by the selected application framework.
- Network: HTTP during local development and HTTPS for deployed environments.
- Operating system: Windows or Linux development/server environment.

2.5 Constraints

- The implementation shall use only one of the permitted stacks: Python (Django/Flask) or PHP (Laravel).
- Only authenticated users may access protected HMS functions.
- A student shall have no more than one active room allocation at a time.
- A room shall not be allocated beyond its configured capacity.
- Only authorized roles may create or modify student, hostel, room, or allocation records.
- The project scope is limited to managing student details and hostel room allocation; unrelated hostel services are excluded.

2.6 Assumptions and dependencies

- Student information entered by authorized staff is assumed to be valid unless flagged or corrected.
- Hostel and room capacity data is entered and maintained by authorized staff.
- Users have access to a supported web browser and network connection for normal operation.
- The selected application framework provides standard authentication/session facilities or equivalent mechanisms.
- Database backups, if required during deployment, are handled by the hosting/development environment.

3. External interface requirements

3.1 User interfaces

The system shall provide a login page, role-specific dashboard, navigation menu, data-entry forms, record tables, search/filter controls, allocation controls, confirmation messages, validation messages, and clear error messages. Student users shall see only their permitted information, while staff users shall see management functions permitted by their roles.

3.2 Hardware interfaces

No specialized hardware interface is required for the core project. The HMS operates on standard computers or mobile devices capable of running a supported web browser.

3.3 Software interfaces

- Web browser for the client interface.
- One of the permitted web application frameworks: Django, Flask, or Laravel.
- Relational database accessed through the selected framework's database facilities or ORM.
- Standard server/runtime environment required by the selected framework.

3.4 Communications

Client-server communication shall use HTTP(S). Deployed systems shall use HTTPS to protect credentials and sensitive student information. Requests that create, update, allocate, transfer, or vacate records shall receive controlled responses and appropriate validation feedback.

4. System features (detailed)

The following requirements use IDs HMS-F-###. Every requirement uses the "shall" form and includes a type, priority, stakeholder/source, acceptance criterion, test-case reference, and dependency/comment so that each requirement can be verified and traced.

4.1 Authentication and Authorization

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-001	The system shall authenticate users using a unique username/email and password before granting access to protected functions.	Functional	High	Student / Warden / Admin	Valid credentials create an authenticated session; invalid credentials are rejected. Test: TC-AUTH-01	Requires user account.
HMS-F-002	The system shall restrict functions and records according to the user role (Student, Warden, Administrator).	Functional	High	Security / Admin	A student cannot access staff-only management functions or unrelated student records. Test: TC-AUTH-02	Depends on RBAC.
HMS-F-003	The system shall provide logout that invalidates the active authenticated session.	Functional	Medium	Student / Warden / Admin	After logout, protected pages require authentication again. Test: TC-AUTH-03	Requires session management.

4.2 Student Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-004	The system shall allow authorized staff to create a student record containing at minimum student ID, name, contact details, course/program, year, and status.	Functional	High	Warden / Admin	Valid student form creates one unique student record visible in staff views. Test: TC-STU-01	Student ID shall be unique.
HMS-F-005	The system shall allow a student to view permitted personal and hostel information associated with the	Functional	High	Student	Logged-in student sees their own permitted details and current	Requires student-account mapping.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	logged-in student account.				allocation information. Test: TC-STU-02	
HMS-F-006	The system shall allow authorized staff to update and deactivate student records without deleting historical allocation records.	Functional	High	Warden / Admin	Edited student values persist; deactivation does not delete historical allocations. Test: TC-STU-03	Historical data shall be preserved.
HMS-F-007	The system shall allow authorized staff to search and filter student records using supported fields such as student ID or name.	Functional	Medium	Warden / Admin	Search/filter returns only matching student records and handles no-match cases. Test: TC-STU-04	Depends on student data.

4.3 Hostel and Room Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-008	The system shall allow authorized staff to create and edit hostel records including hostel name, block/location, and status.	Functional	High	Warden / Admin	Valid hostel data is stored and displayed in hostel lists. Test: TC-ROOM-01	Hostel identifier shall be unique.
HMS-F-009	The system shall allow authorized staff to create and edit room records containing hostel, room number, room type, capacity, and status.	Functional	High	Warden / Admin	Room appears under the selected hostel with the configured capacity and status. Test: TC-ROOM-02	Room number unique within hostel.
HMS-F-010	The system shall display room availability as vacant, partially occupied, or full based on active	Functional	High	Warden / Admin	Availability label and occupied count match active allocation	Depends on allocation data.

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	allocations and configured capacity.				records. Test: TC-ROOM-03	

4.4 Allocation and Transfer Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-011	The system shall prevent allocation of a room when its active occupancy has reached configured capacity.	Functional	High	Business / Warden	Allocation attempt at full capacity is rejected and no allocation record is created. Test: TC-ROOM-04	Capacity validation required.
HMS-F-012	The system shall allow authorized staff to allocate an eligible student to an available room and store the allocation date and active status.	Functional	High	Warden / Admin	Valid allocation creates one active allocation and room occupancy increases by one. Test: TC-ALLOC-01	Student must be active and room available.
HMS-F-013	The system shall prevent a student from having more than one active room allocation at the same time.	Functional	High	Business / Warden	Allocation attempt for a student with an active allocation is rejected. Test: TC-ALLOC-02	Requires active-allocation rule.
HMS-F-014	The system shall allow authorized staff to transfer a student from one room to another while closing the previous active allocation and creating the new active allocation.	Functional	High	Warden / Admin	After transfer, previous allocation is inactive and the new room has one additional active occupant. Test: TC-ALLOC-03	Both rooms must satisfy capacity rules.

4.5 Allocation Records and Reporting

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-015	The system shall allow authorized staff to vacate a student allocation by recording the checkout/end date and changing the allocation to inactive.	Functional	High	Warden / Admin	Vacating ends the active allocation and decreases room occupancy by one. Test: TC-ALLOC-04	Requires active allocation.
HMS-F-016	The system shall allow students and authorized staff to view the current room allocation details, including hostel, room number, allocation date, and allocation status, according to role permissions.	Functional	High	Student / Warden / Admin	Displayed allocation details match the active allocation record. Test: TC-ALLOC-05	Depends on allocation record.
HMS-F-017	The system shall allow authorized staff to view a student's allocation history, including previous rooms and allocation start/end dates.	Functional	Medium	Warden / Admin	Historical allocations remain visible after transfer or vacating and are ordered consistently. Test: TC-ALLOC-06	Requires preserved history.
HMS-F-018	The system shall provide an authorized staff view summarizing hostel room occupancy and current allocation counts.	Functional	Medium	Warden / Admin	Summary counts equal the number of active allocations grouped by hostel/room. Test: TC-REPORT-01	Read-only summary derived from current data.

5. Non-functional requirements (detailed)

The template requires at least five non-functional requirements. The following seven requirements are measurable and are tied to verification methods applicable to the HMS mini-project.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HMS-NF-001	For normal project test load, 95% of ordinary authenticated page requests shall complete within 2 seconds, excluding file downloads.	Performance	High	Measured over at least 50 representative requests; 95th percentile ≤ 2 s. Test: TC-NFR-01
HMS-NF-002	The deployed system shall achieve at least 99% availability during the evaluation period, excluding planned maintenance.	Reliability	Medium	Availability log shows $\geq 99\%$ during the evaluation period. Test: TC-NFR-02
HMS-NF-003	The application shall preserve consistent allocation data so that no student has more than one active allocation and no room exceeds configured capacity.	Data Integrity	High	Verification queries show zero duplicate-active-student or over-capacity violations after allocation tests. Test: TC-NFR-03
HMS-NF-004	The primary student and staff workflows shall be usable through a current desktop browser without horizontal scrolling at a viewport width of 1280 px.	Usability	Medium	Reviewer completes login, student lookup, room lookup, and allocation workflows without layout overflow. Test: TC-NFR-04
HMS-NF-005	The application code shall be organized into clear presentation, application/business, and data-access responsibilities using the selected project framework.	Maintainability	Medium	Code review confirms reasonable separation of models/data, views/controllers, templates/UI, forms, and configuration. Test: TC-NFR-05

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HMS-NF-006	The system shall return a controlled error response for invalid input, missing records, and unauthorized access without exposing stack traces to end users in production-like settings.	Robustness	High	Negative test suite produces controlled error messages/pages and no debug traceback. Test: TC-NFR-06
HMS-NF-007	The user interface shall provide labels, visible validation messages, and keyboard-accessible form controls for all core student, room, and allocation workflows.	Accessibility / Usability	Medium	Manual accessibility checklist passes for core forms and navigation. Test: TC-NFR-07

5.1 Security

5.1.1 Security Objectives

SO-1 Confidentiality and least privilege: protect student personal information, credentials, and administrative records so that users can access only information and functions authorized for their role.

SO-2 Integrity and accountability: prevent inconsistent or unauthorized changes to student, room, and allocation records and retain sufficient audit information to identify critical administrative changes.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
HMS-SR-001	The system shall store user passwords only as one-way password hashes using the selected framework's approved password-hashing mechanism; plaintext passwords shall not be stored.	Security	High	Code/configuration inspection and authentication test pass. Test: TC-SEC-01
HMS-SR-002	The system shall enforce role-based authorization on every protected staff function and shall reject unauthorized direct URL access.	Security	High	Student role attempts staff URL; request is denied and no protected data is returned. Test: TC-SEC-02
HMS-SR-003	The system shall terminate or require re-authentication after 30 minutes of inactivity for authenticated sessions in the deployed configuration.	Security	Medium	Session created, left inactive, then protected request requires

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
				authentication. Test: TC-SEC-03
HMS-SR-004	The deployed system shall use HTTPS for authenticated traffic and shall enable secure session/cookie settings appropriate to HTTPS deployment.	Security	High	HTTPS is enforced and authenticated session cookie is marked secure in deployment configuration. Test: TC-SEC-04
HMS-SR-005	The system shall record timestamp, user identity, action type, and affected record identifier for critical student, room, allocation, and role changes; passwords shall never be recorded in audit data.	Security	High	Audit log contains required fields for test operations and contains no password values. Test: TC-SEC-05

6. Quality attributes & Acceptance tests

6.1 Quality attributes

Attribute	Target	Verification
Performance	95% of ordinary page requests ≤ 2 s under normal project test load	TC-NFR-01
Reliability	$\geq 99\%$ availability for the evaluation environment, planned maintenance excluded	TC-NFR-02
Data integrity	No duplicate active student allocation and no room over-capacity after tests	TC-NFR-03
Usability	Core workflows usable at 1280 px width without horizontal scrolling	TC-NFR-04
Maintainability	Clear separation of application responsibilities and readable naming	TC-NFR-05
Robustness	No raw stack traces/debug pages exposed for invalid/unauthorized requests	TC-NFR-06
Accessibility	Labels and keyboard-accessible controls for core forms	TC-NFR-07

6.2 Acceptance test suites

Test suite	Area	Acceptance activity	Requirements
TC-AUTH-01	Authentication	Valid/invalid login; authenticated session created only for valid credentials.	HMS-F-001
TC-AUTH-02	Authorization	Student, Warden, and Administrator access boundaries verified.	HMS-F-002
TC-AUTH-03	Logout	Logout invalidates session and protected page access fails.	HMS-F-003
TC-STU-01..04	Student management	Create, view, update/deactivate, search and filter student records.	HMS-F-004..007
TC-ROOM-01..04	Hostel/room	Create hostels/rooms, show availability, and block over-capacity allocation.	HMS-F-008..011
TC-ALLOC-01..06	Room allocation	Allocate, prevent duplicate active allocation, transfer, vacate, view current details, and verify history.	HMS-F-012..017
TC-REPORT-01	Occupancy summary	Verify current occupancy and allocation counts by hostel/room.	HMS-F-018
TC-NFR-01..07	Non-functional	Performance, availability, integrity, usability, maintainability, robustness, accessibility checks.	HMS-NF-001..007
TC-SEC-01..05	Security	Password hashing, RBAC, session timeout, HTTPS/cookies, audit logging.	HMS-SR-001..005

6.3 Acceptance exit criteria

- All High-priority functional requirements are implemented and pass their acceptance tests.
- All five security requirements are implemented and pass security acceptance testing.
- All mandatory non-functional requirements have measurable evidence or documented evaluation results.
- No critical defect remains in authentication, authorization, student-data handling, room-capacity enforcement, or allocation integrity.
- The RTM contains a test reference for every requirement and final status is updated after verification.

7. System models and diagrams

7.1 Student use-case diagram

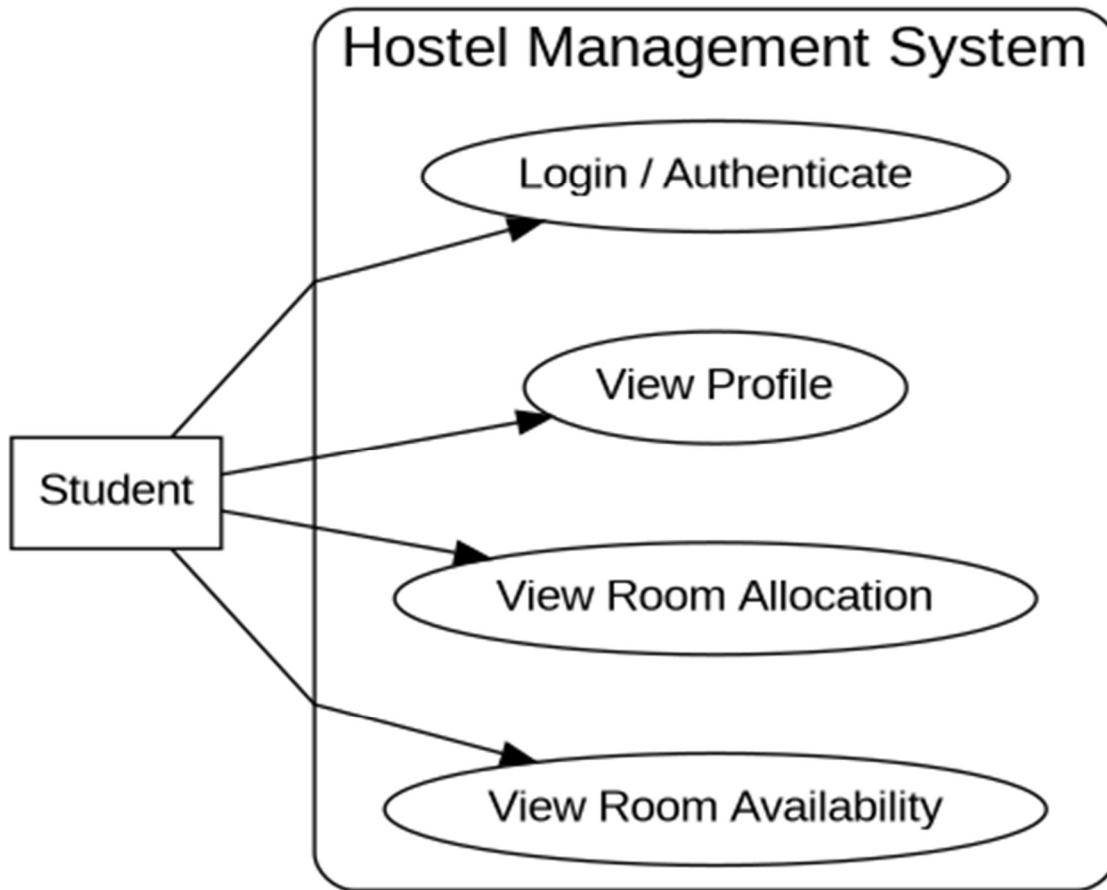


Figure 2. UML-style use-case diagram - Student interactions

7.2 Warden and administrator use-case diagram

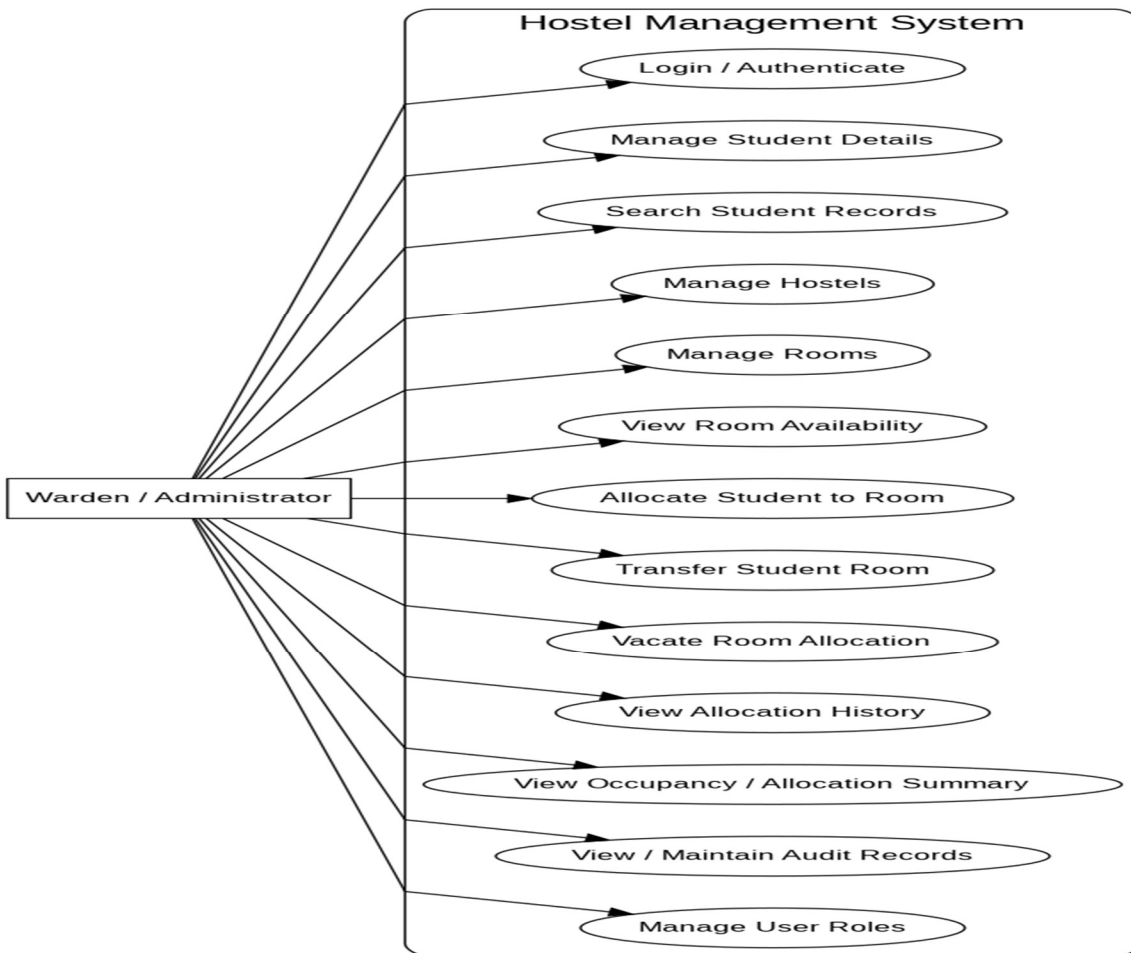


Figure 3. UML-style use-case diagram - Warden and administrator interactions

7.3 High-level data model

The core persistent entities are User, Student, Hostel, Room, and Allocation, with AuditLog supporting accountability. A Student may have many historical Allocation records but at most one active allocation. A Hostel contains many Rooms, and a Room has many Allocation records over time.

Entity	Key information	Relationship summary
User	id, username/email, password hash, role, active	One user maps to one application role; staff actions create audit records.
Student	student_id, name, contact, course/program, year, status	One student has many historical allocations and at most one active allocation.
Hostel	hostel_id, name, block/location, status	One hostel contains many rooms.

Entity	Key information	Relationship summary
Room	room_id, hostel, room_no, type, capacity, status	One room has many allocation records over time.
Allocation	allocation_id, student, room, start/end date, active status	Links students to rooms and preserves allocation history.
AuditLog	timestamp, user, action, entity, record id	Captures critical student, room, allocation, and role changes.

8. Requirements Traceability Matrix (RTM)

The RTM links every functional, non-functional, and security requirement to its SRS section, design reference, implementation module, test case, and current verification status. Status values are N = Not yet verified, P = Passed, and A = Accepted/approved.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
HMS-F-001	Authenticate user	4.1 / DS-AUTH-01	Authentication	TC-AUTH-01	N	
HMS-F-002	Enforce role-based access	4.1 / DS-AUTH-01	Authentication	TC-AUTH-02	N	
HMS-F-003	Logout user	4.1 / DS-AUTH-01	Authentication	TC-AUTH-03	N	
HMS-F-004	Create student record	4.2 / DS-STU-01	Student	TC-STU-01	N	
HMS-F-005	View own student details	4.2 / DS-STU-01	Student	TC-STU-02	N	
HMS-F-006	Update/deactivate student	4.2 / DS-STU-01	Student	TC-STU-03	N	
HMS-F-007	Search student records	4.2 / DS-STU-01	Student	TC-STU-04	N	
HMS-F-008	Manage hostel records	4.3 / DS-ROOM-01	Hostel/Room	TC-ROOM-01	N	
HMS-F-009	Manage room records	4.3 / DS-ROOM-01	Hostel/Room	TC-ROOM-02	N	
HMS-F-010	Show room availability	4.3 / DS-ROOM-01	Hostel/Room	TC-ROOM-03	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
HMS-F-011	Prevent over-capacity allocation	4.3 / DS-ROOM-01	Hostel/Room	TC-ROOM-04	N	
HMS-F-012	Allocate student to room	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-01	N	
HMS-F-013	Prevent duplicate active allocation	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-02	N	
HMS-F-014	Transfer student room	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-03	N	
HMS-F-015	Vacate room allocation	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-04	N	
HMS-F-016	View current allocation	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-05	N	
HMS-F-017	View allocation history	4.4 / DS-ALLOC-01	Allocation	TC-ALLOC-06	N	
HMS-F-018	View occupancy summary	4.5 / DS-REPORT-01	Reporting	TC-REPORT-01	N	
HMS-NF-001	Performance target	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-01	N	
HMS-NF-002	Availability target	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-02	N	
HMS-NF-003	Allocation data integrity	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-03	N	
HMS-NF-004	Usable responsive layout	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-04	N	
HMS-NF-005	Maintainable architecture	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-05	N	
HMS-NF-006	Controlled error handling	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-06	N	
HMS-NF-007	Accessible core forms	5 / DS-NFR-01	Quality / Cross-cutting	TC-NFR-07	N	
HMS-SR-001	Password hashing	5.1.2 / DS-SEC-01	Security	TC-SEC-01	N	

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
HMS-SR-002	Role-based authorization	5.1.2 / DS-SEC-01	Security	TC-SEC-02	N	
HMS-SR-003	Session timeout	5.1.2 / DS-SEC-01	Security	TC-SEC-03	N	
HMS-SR-004	HTTPS and secure cookies	5.1.2 / DS-SEC-01	Security	TC-SEC-04	N	
HMS-SR-005	Audit logging	5.1.2 / DS-SEC-01	Security	TC-SEC-05	N	